

HAT-P-12b

R-0772

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-12_230503 · created 2026-07-28T05:22:56+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460067.7446 +/- 0.0042 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.139 +/- 0.046
Transit depth (Rp/Rs)^2	1.94 +/- 1.3 [%]
Orbital Inclination (inc)	84.65 +/- 1.82
Airmass coefficient 1 (a1)	0.922 +/- 0.025
Airmass coefficient 2 (a2)	0.066 +/- 0.04
Scatter in the residuals of the lightcurve fit is	2.66 %
Best Comparison Star	#2 - [422, 164]
Optimal Method	PSF photometry
Transit Duration (day)	0.034 +/- 0.035

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.139 ± 0.046 vs 0.1406 (deviation 0.03σ)
Duration fit vs published	0.034 d vs 0.0975 d (deviation 1.36σ)
Mid-time O-C	-10.5 min
Depth SNR	2.3
Residual scatter	2.66 %

Light curve

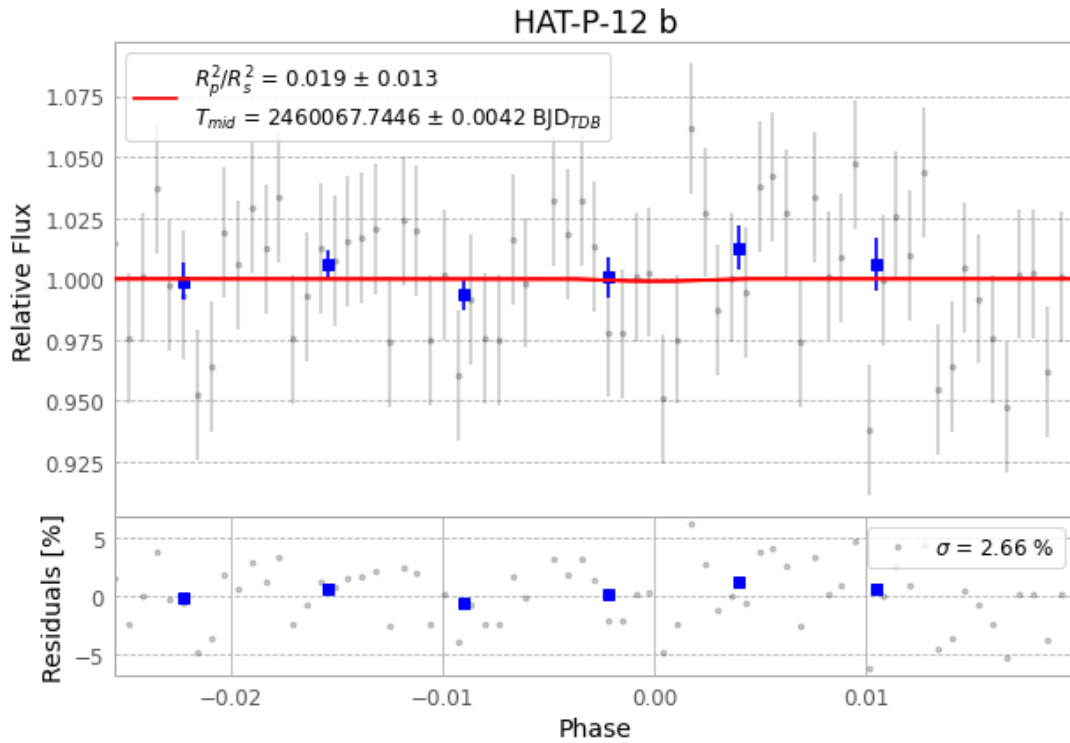


Figure 1 — FinalLightCurve_HAT-P-12 b_2023-05-02.png (SHA-256 673f224bf77f1d6b...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [353, 275], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 1d0b04ad39abcdce...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

71 FITS frames (47 MB), HATP-12230503034815.FITS → HATP-12230503071815.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-12-230503.log (cfb31214740b...), FinalLightCurve_HAT-P-12 b_2023-05-02.png (673f224bf77f...), FinalLightCurve_HAT-P-12 b_2023-05-02.csv (048e9e17dfc1...)

Compute resources

Wall time 160.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-28 05:22Z.